Rishie Raj

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Experience

UMD Computer Science Dept. **College Park, MD**
AI Researcher — Multimodal Systems *Apr 2025 - Present*

- Researcher in GAMMA Lab at UMD, working on data collection, dataset modeling, experimental evaluation, and large-scale analysis for audio-visual large language models, multimodal AI systems, and AI agent frameworks.
- Collaborating with Tianyi Lab at UMD to formulate research problems, develop and prototype image editing AI agents, and conduct controlled experiments for agentic decision-making and multimodal reasoning.
- Trained and fine-tuned large multimodal models on GPU clusters, optimizing throughput and memory utilization.

Maruti Suzuki India Ltd. (MSIL) **Gurugram, India**
Product Manager — R&D *Jul 2018 - Apr 2021*

- Worked as project manager on large-scale, cross-functional engineering teams, leading the Ertiga and its variants, India's highest-selling MPV, involving data-driven decision-making and stakeholder communication.
- Contributed to diesel BS-6/EURO-6 powertrain development, MSIL's first self-designed diesel engine, applying engineering experimentation, validation, and performance analysis under regulatory constraints.
- Implemented a cost-optimization initiative (∼2,500 INR/vehicle) through quantitative analysis of materials and manufacturing trade-offs, directly supporting productivity and commercial goals.
- Proposed a process optimization layout for the Weld shop ventilation system using operational data analysis, reducing wastage by ∼30% and improving system efficiency.
- Played a key role in eliciting customer and market requirements and translating them into well-defined product specifications, enabling data-informed design decisions aligned with business objectives.
- Led schedule and milestone prioritization across multi-disciplinary design and development teams, ensuring stakeholder alignment, timely execution, and efficient resource utilization.

Publications

1. **FaSTA*: Fast-Slow Toolpath Agent with Subroutine Mining for Efficient Multi-turn Image Editing**
   Advait Gupta, *Rishie Raj*, Dang Nguyen, Tianyi Zhou
   ICLR 2026
2. **Foundations and Frontiers of Multimodal Agentic Frameworks**
   Neel Mokaria*, *Rishie Raj**, Dheeraj Baiju*, Xiaoqian Shen, Shraman Pramanick, Kevin Qinghong Lin, Mike Zheng Shou, Philip Torr, Mohamed Elhoseiny, Yapeng Tian, Ruohan Gao, Salman Khan, Sayan Nag, Sanjoy Chowdhury, Dinesh Manocha
   Under Review
3. **M^3-Edit: An Agentic Framework for Multimodal, Multireference, Multitask Editing**
   Advait Gupta, *Rishie Raj*, Ruohan Gao
   Under Review
4. **AURA: A Fine-Grained Benchmark and Decomposed Metric for Audio-Visual Reasoning**
   Siminfar Samakoush Galougah*, *Rishie Raj**, Sanjoy Chowdhury, Sayan Nag, Ramani Duraiswami
   Under Review

Projects

Multi-modal Agent for Reference-Guided Image Editing *Apr 2025*

- Developed an LLM-based AI agent that performs complex, multi-step image edits using indirect multimodal signals, demonstrating hypothesis-driven agent behavior without task-specific training.
- Proposed a two-stage planning and execution framework that improves flexibility by decoupling reference analysis from action execution, aligning with modern AI agent architectures.

Detecting AI-Generated Images in Zero-Shot Setting *Dec 2024*

- Proposed a novel statistical framework by adapting perplexity and cross-perplexity metrics for evaluation of image generative models.
- Achieved robust, training-free detection across multiple generative models (Stable Diffusion, DALL·E, Emu3), demonstrating generalization and reproducibility.

Object Detection in Adverse Weather Conditions *May 2024*

- Designed a deep learning-based CNN defogging model, predicting parameters for differentiable functions via learned representations.
- Integrated the system with YOLOv5 for real-time object detection, improving mAP through systematic experimentation and benchmarking.

Real Time Semantic Segmentation using Efficient Neural Network *May 2024*

- Developed a lightweight CNN encoder-decoder architecture for semantic segmentation that is optimized for real-time inference and deployment constraints.
- Achieved strong segmentation accuracy while maintaining computational efficiency through architectural optimization.

Project NavGuide: A Swarm Navigation Robot System for Warehouse Applications *Dec 2024*

- Designed and implemented a multi-agent robotic swarm system, leveraging the RVO2 obstacle avoidance algorithm to model decentralized agent coordination and collision avoidance.
- Conducted simulation-based experimentation in Gazebo using a five-agent TurtleBot3 swarm, evaluating system behavior in a structured warehouse environment.

Education

University of Maryland, College Park **College Park, MD**
M.Eng. in Robotics; ***GPA: 4.0/4.0*** *Aug 2023 - May 2025*

Odisha University of Technology and Research **Bhubaneswar, India**
B.Tech in Mechanical Engineering; ***GPA: 8.0/10.0*** *Aug 2014 - May 2018*

Technical Skills

Languages: Python, C++
Tools: GitHub, GCP, Docker, ROS, RViz, MATLAB, LaTeX, CUDA
Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, Pandas, Matplotlib, Hugging Face, LangGraph
Concepts: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Cloud Computing, Object-Oriented Programming, Software Development